

VMware vSAN

Certification Interview Preparation

300 Expert Questions & Answers

Fundamentals • Architecture • Security • ESA • Troubleshooting

VCP-DCV | VCAP-DCA | VCAP-DCD | VCDX

Covers vSAN 6.x through vSAN 8 ESA

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◆ 1. vSAN Fundamentals & Architecture

15 Questions – Covering key concepts and exam topics

Q1. What is VMware vSAN?

► **Answer:** VMware vSAN is a software-defined storage solution that pools direct-attached storage devices (HDDs, SSDs, NVMe) across ESXi hosts to create a shared datastore. It is natively integrated into the ESXi hypervisor and managed through vCenter Server.

Q2. What are the key components of vSAN architecture?

► **Answer:** vSAN consists of disk groups (cache + capacity tiers), the vSAN cluster (ESXi hosts), the vSAN datastore (distributed object store), and the vSAN network. Each host contributes storage devices to form a hyper-converged infrastructure.

Q3. What is a disk group in vSAN?

► **Answer:** A disk group is the fundamental storage unit in vSAN. It consists of one cache device (SSD/NVMe) and one to seven capacity devices (HDD or SSD). Each ESXi host can have up to five disk groups.

Q4. What is the difference between Hybrid and All-Flash vSAN?

► **Answer:** Hybrid vSAN uses SSDs for caching and HDDs for capacity. All-Flash vSAN uses SSDs/NVMe for both cache and capacity tiers. All-Flash provides better performance and lower latency. In All-Flash, the cache device is used exclusively for write buffering.

Q5. What is vSAN ESA (Express Storage Architecture)?

► **Answer:** vSAN ESA, introduced in vSAN 8, is a next-generation architecture that utilizes a single-tier storage model using NVMe devices only. It eliminates the traditional cache/capacity tier split and offers improved performance, compression, and RAID-5/6 efficiency.

Q6. How many hosts are required for a vSAN cluster?

► **Answer:** A minimum of 3 hosts is required for a standard vSAN cluster to tolerate one host failure (FTT=1 with RAID-1). For stretched clusters, a minimum of 2 data site hosts plus 1 witness is needed.

Q7. What is the maximum number of hosts in a vSAN cluster?

► **Answer:** A vSAN cluster supports a maximum of 96 hosts per cluster (64 in older versions prior to vSAN 7). This limit applies to both standard and stretched cluster configurations.

Q8. What is the vSAN witness appliance?

► **Answer:** The vSAN witness is a lightweight VM deployed at a third site (or separate location) in stretched and two-node configurations. It holds witness components (metadata) to maintain quorum and prevent split-brain scenarios without consuming capacity.

Q9. What network ports does vSAN use?

► **Answer:** vSAN requires a dedicated VMkernel port (vmk) for vSAN traffic on each host. The recommended bandwidth is 10GbE or higher. vSAN uses TCP/IP for cluster communication on port 2233 and multicast for initial discovery (though unicast is supported in newer versions).

Q10. What is the vSAN Object Store?

► **Answer:** vSAN uses a distributed object store where all VM data (VMDKs, swap files, snapshots) are stored as objects. Each object is broken into components distributed across hosts according to the storage policy.

Q11. What is a vSAN component?

► **Answer:** A component is the smallest unit of storage in vSAN. Each object (e.g., VMDK) is split into components distributed across hosts. Components include data components, witness components, and parity components depending on the RAID configuration.

Q12. What is a vSAN stripe?

► **Answer:** Striping in vSAN distributes data across multiple capacity devices for improved performance. The Number of Disk Stripes per Object policy setting controls how many stripes are created (1–12). More stripes mean better I/O parallelism.

Q13. What is the vSAN CLOM?

► **Answer:** CLOM (Cluster Level Object Manager) is responsible for placement and rebalancing of vSAN objects across the cluster. It monitors cluster health and triggers resyncing when a host or disk failure occurs.

Q14. What is DOM in vSAN?

► **Answer:** DOM (Distributed Object Manager) is the component that handles I/O operations for vSAN objects. It consists of DOM Owner (coordinates I/O) and DOM Client (sends I/O from VM) working together across the cluster.

Q15. What is LSOM in vSAN?

► **Answer:** LSOM (Local Storage Object Manager) is responsible for managing physical disk access on each ESXi host. It handles reads, writes, and the log-structured file system within each disk group.

◆ 2. Storage Policies & Compliance

15 Questions – Covering key concepts and exam topics

Q16. What is a vSAN Storage Policy?

► **Answer:** A vSAN Storage Policy (SPBM - Storage Policy Based Management) defines the level of protection, performance, and availability for VM storage objects. Policies are assigned per VM or per VMDK and are enforced by the vSAN cluster.

Q17. What does FTT mean in vSAN?

► **Answer:** FTT (Failures To Tolerate) defines how many host, disk, or network failures a VM can withstand without data loss. FTT=1 tolerates 1 failure, requiring at least 3 hosts. FTT=2 tolerates 2 failures, requiring at least 5 hosts. FTT=3 requires 7 hosts.

Q18. What is the difference between RAID-1 and RAID-5/6 in vSAN?

► **Answer:** RAID-1 (Mirroring) creates full copies of data across hosts and requires more raw capacity. RAID-5 (erasure coding) uses parity and requires a minimum of 4 hosts with ~33% overhead. RAID-6 uses dual parity with a minimum of 6 hosts and ~50% overhead but tolerates 2 failures.

Q19. What is the PFTT policy?

► **Answer:** PFTT (Primary Level of Failures to Tolerate) is the top-level failure tolerance for stretched clusters. It controls how many site-level failures can be tolerated, typically set to 1 for stretched cluster configurations with two data sites.

Q20. What happens when a VM is non-compliant in vSAN?

► **Answer:** A non-compliant VM means its storage objects do not meet the assigned storage policy. This can occur due to host failures, insufficient capacity, or policy changes. vSAN will attempt to restore compliance automatically when resources are available.

Q21. How do you apply a storage policy to a VM in vSAN?

► **Answer:** You can apply a storage policy by right-clicking the VM in vCenter > Edit Settings > VM Storage Policy, or during VM deployment by selecting the policy on the storage page. Policies can be applied per VM or per individual VMDK.

Q22. What is the default vSAN storage policy?

► **Answer:** The default vSAN storage policy is 'vSAN Default Storage Policy' which uses FTT=1 with RAID-1 mirroring. This requires a minimum of 3 hosts and creates 2 copies of data with 1 witness component.

Q23. What does 'Object Space Reservation' mean in vSAN policy?

► **Answer:** Object Space Reservation (OSR) specifies what percentage of a virtual disk's logical size is reserved as thick-provisioned. A value of 0% means thin provisioning; 100% means fully thick provisioned at the object level.

Q24. What is the 'Flash Read Cache Reservation' policy?

► **Answer:** Flash Read Cache Reservation reserves a portion of the SSD cache for read caching for a specific VM or VMDK. It is expressed as a percentage of the logical disk size. This is mainly applicable in Hybrid vSAN environments.

Q25. What is Force Provisioning in vSAN?

► **Answer:** Force Provisioning allows a VM to be deployed even if the cluster cannot currently satisfy the assigned storage policy. The object is marked non-compliant but the VM can still run. This is useful during maintenance or temporary resource shortfalls.

Q26. How many components can a single vSAN object have?

► **Answer:** A single vSAN object can have a maximum of 32 components. Large VMDKs (over 255GB) are automatically split into multiple objects. Each object follows the storage policy independently.

Q27. What is IOPS Limit for Object in vSAN policy?

► **Answer:** The IOPS Limit policy caps the maximum IOPS a VM's storage object can consume. This is useful for QoS (Quality of Service) to prevent noisy neighbor issues in multi-tenant environments. A value of 0 means no limit.

Q28. What is the Checksum Disabled policy?

► **Answer:** The Checksum Disabled policy turns off end-to-end data integrity checking for specific objects. By default, vSAN performs checksums on all data. Disabling it can slightly improve performance but reduces data integrity validation.

Q29. What are VM storage policies used for beyond vSAN?

► **Answer:** While SPBM was originally created for vSAN, it can also apply to external storage like VVOL (Virtual Volumes) arrays, ensuring consistent policy-driven storage management across different backends through a unified framework.

Q30. How does vSAN handle policy changes on running VMs?

► **Answer:** When you change a storage policy on a running VM, vSAN evaluates compliance and triggers a background resync process to restructure the storage objects to meet the new policy. The VM remains online during this process.

◆ 3. vSAN Networking

10 Questions — Covering key concepts and exam topics

Q31. What are the network requirements for vSAN?

► **Answer:** vSAN requires a dedicated VMkernel port (vmk) tagged for vSAN traffic on each ESXi host. Minimum bandwidth is 1GbE for hybrid and 10GbE for All-Flash. Low latency (< 5ms RTT) between hosts is recommended.

Q32. Does vSAN support jumbo frames?

► **Answer:** Yes, vSAN supports jumbo frames (MTU 9000). Enabling jumbo frames can improve performance by reducing CPU overhead and increasing throughput. All switches and host vNICs in the vSAN path must be configured with matching MTU.

Q33. What is the recommended vSAN network topology?

► **Answer:** VMware recommends a dedicated, non-routed Layer 2 network for vSAN traffic. For stretched clusters, a Layer 3 routed network between sites is required with RTT < 5ms.

Q34. Can vSAN use multicast or unicast?

► **Answer:** Early versions of vSAN required multicast for cluster discovery. From vSAN 6.6 onward, unicast-only mode is supported and preferred. Unicast simplifies network configuration and removes multicast switch dependencies.

Q35. What is the impact of network latency on vSAN?

► **Answer:** High network latency negatively impacts vSAN performance. VMware recommends maximum 1ms RTT between hosts within a single site, and maximum 5ms RTT for stretched cluster inter-site links. Exceeding these values can cause I/O timeouts and VM stalls.

Q36. How does vSAN handle network partitions?

► **Answer:** When a network partition occurs, vSAN uses quorum-based algorithms to determine which partition retains data access. The partition with majority components (or witness) remains accessible. VMs on the minority side are paused to prevent split-brain data corruption.

Q37. What is vSAN Data-in-Transit Encryption?

► **Answer:** vSAN Data-in-Transit Encryption (vSAN 6.7.1+) encrypts data as it travels across the vSAN network between hosts using AES-128-GCM. This protects against network eavesdropping without requiring a KMS (Key Management Server).

Q38. How do you configure vSAN VMkernel adapter?

► **Answer:** In vCenter, navigate to Host > Configure > VMkernel Adapters, add a new VMkernel adapter, assign an IP address, and enable the vSAN service checkbox. Alternatively use esxcli or PowerCLI to configure it programmatically.

Q39. What is the vSAN health check for network?

► **Answer:** vSAN health checks include tests for MTU mismatch, network latency, packet loss, and vSAN VMkernel adapter connectivity. These can be found in vCenter under Cluster > Monitor > vSAN > Skyline Health.

Q40. Can vSAN share a VMkernel port with vMotion?

► **Answer:** While technically possible to share a VMkernel port, VMware does not recommend it. Mixing vSAN and vMotion traffic can cause performance degradation during migrations. Best practice is to use dedicated NICs for vSAN traffic.

◆ 4. Fault Domains & Stretched Clusters

10 Questions — Covering key concepts and exam topics

Q41. What is a vSAN Fault Domain?

► **Answer:** A Fault Domain is a logical grouping of ESXi hosts that represent a potential single point of failure (e.g., a rack or chassis). vSAN distributes object components across different fault domains to improve resilience against rack-level failures.

Q42. What is a vSAN Stretched Cluster?

► **Answer:** A vSAN Stretched Cluster spans two geographically separate sites (data sites) plus a third witness site. It provides site-level high availability and disaster avoidance, allowing VMs to continue running even if an entire site fails.

Q43. What is a two-node vSAN cluster?

► **Answer:** A two-node vSAN cluster uses two ESXi hosts with a separate witness appliance. It is designed for remote office/branch office (ROBO) environments where full cluster hardware is not economical. Minimum bandwidth between nodes is 1GbE.

Q44. What is the role of a vSAN witness in stretched clusters?

► **Answer:** The witness host stores only witness components (metadata/tiebreaker) for vSAN objects. It participates in quorum decisions but does not store actual VM data. It must be on a third site separate from both data sites.

Q45. How many hosts per site are required for a stretched cluster?

► **Answer:** A minimum of 2 hosts per data site (4 total data hosts + 1 witness) is required for a stretched cluster. Each site should have equal capacity and resources for balanced VM placement.

Q46. What is the preferred site in a stretched cluster?

► **Answer:** The preferred site is the primary data site in a stretched cluster. In a split-brain scenario where both sites have equal votes, vSAN favors the preferred site for continued operation. The preference is set during stretched cluster configuration.

Q47. What is the RTT requirement for stretched cluster inter-site links?

► **Answer:** VMware requires maximum 5ms RTT between data sites and maximum 200ms RTT between data sites and the witness site. Exceeding these values can cause cluster instability and I/O failures.

Q48. How does vSAN handle a full site failure in a stretched cluster?

► **Answer:** When a site fails, vSAN detects the failure through heartbeat timeouts. The surviving site (with majority quorum via witness) takes over. VMs on the failed site are restarted on the surviving site using VM-level HA, resulting in RTO measured in minutes.

Q49. What is vSAN Local Protection policy in stretched clusters?

► **Answer:** The SFTT (Secondary Failures to Tolerate) policy provides protection within each site of a stretched cluster. SFTT=1 with RAID-1 maintains two copies within each site, protecting against individual host failures within the site.

Q50. Can fault domains be used with All-Flash vSAN?

► **Answer:** Yes, fault domains are supported in both Hybrid and All-Flash vSAN configurations. Fault domains require a minimum of 4 fault domains to support FTT=1 with RAID-5 erasure coding.

◆ 5. Encryption & Security

10 Questions – Covering key concepts and exam topics

Q51. What types of encryption does vSAN support?

► **Answer:** vSAN supports Data-at-Rest Encryption (using a KMS) and Data-in-Transit Encryption (built-in, no KMS required). Both can be used simultaneously for maximum security. Encryption is configured at the cluster level.

Q52. What is required for vSAN Data-at-Rest Encryption?

► **Answer:** Data-at-Rest Encryption requires: a compatible KMIP 1.1 Key Management Server (KMS), vCenter Server with KMS trust established, vSAN All-Flash cluster (recommended), and ESXi 6.6+ hosts. The KMS manages AES-256 encryption keys.

Q53. How does vSAN key management work?

► **Answer:** vCenter Server communicates with the KMS to obtain Key Encryption Keys (KEKs). Each ESXi host generates a local Data Encryption Key (DEK) that is wrapped by the KEK. The DEK encrypts/decrypts data on disk while the KEK is stored on the KMS.

Q54. What happens to encrypted vSAN data when KMS is unavailable?

► **Answer:** If KMS becomes unavailable, hosts that have already unlocked their DEKs continue operating normally. However, hosts that reboot cannot unlock their DEKs without KMS access and will not be able to mount their disk groups, causing data inaccessibility.

Q55. What is HCI Mesh and how does it relate to security?

► **Answer:** HCI Mesh (vSAN 7.0 U1+) allows a compute cluster to consume storage from a vSAN cluster. Security policies and encryption apply to the storage cluster. Traffic between clusters uses the vSAN network and should be secured accordingly.

Q56. What is vSAN Host Encryption Mode?

► **Answer:** Host Encryption Mode is a prerequisite for vSAN encryption. It must be enabled on each ESXi host before vSAN encryption can be configured. When enabled, all core dumps on the host are also encrypted for security.

Q57. How do you re-key vSAN encryption?

► **Answer:** Re-keying can be performed through vCenter: shallow re-key replaces KEKs from the KMS without re-encrypting data on disk (fast). Deep re-key replaces both KEKs and generates new DEKs, requiring data to be re-encrypted on disk (slower, more disruptive).

Q58. Does vSAN encryption work with deduplication and compression?

► **Answer:** vSAN performs deduplication and compression before encryption. This means dedupe and compression are effective because they operate on plaintext data. Encryption is applied after, ensuring both storage efficiency and security.

Q59. What is FIPS compliance in vSAN?

► **Answer:** vSAN supports FIPS (Federal Information Processing Standard) 140-2 validated cryptographic modules. FIPS mode can be enabled on ESXi hosts and ensures that only FIPS-approved algorithms are used for encryption operations.

Q60. How does vSAN handle certificate management?

► **Answer:** vSAN uses vSphere Certificate Management for internal component communication. VMCA (VMware Certificate Authority) issues certificates to ESXi hosts. Certificates can be replaced with custom CA-signed certificates for compliance requirements.

◆ 6. Deduplication, Compression & Space Efficiency

10 Questions – Covering key concepts and exam topics

Q61. What is vSAN deduplication?

► **Answer:** vSAN deduplication eliminates duplicate data blocks within a disk group to reduce storage consumption. It operates at the block level (4KB block size in OSA) and is performed in the cache tier before data is written to capacity devices.

Q62. What is vSAN compression?

► **Answer:** vSAN compression reduces the size of data blocks before writing them to capacity devices. It uses LZ4 compression algorithm. Compression can be enabled independently of deduplication and works on capacity-tier data.

Q63. Can deduplication and compression be enabled separately?

► **Answer:** Compression can be enabled without deduplication. However, deduplication requires compression to be enabled simultaneously. Both are cluster-wide settings in OSA. In ESA, they are enabled per storage policy.

Q64. What are the minimum hardware requirements for dedupe/compression?

► **Answer:** Deduplication and compression in All-Flash OSA require at least 4 ESXi hosts, NVMe or SSD cache devices, and sufficient memory. VMware recommends 512GB+ raw capacity per disk group for effective space savings.

Q65. What is the dedupe ratio in vSAN?

► **Answer:** The deduplication ratio varies by workload. VDI environments typically see 4:1 to 8:1 ratios. Mixed workloads may see 1.5:1 to 3:1. Database workloads with random/unique data may see minimal deduplication benefits.

Q66. How does vSAN ESA handle space efficiency?

► **Answer:** vSAN ESA uses inline compression at the log-structured level, applying compression before data is written to NVMe devices. ESA achieves higher compression ratios than OSA due to its optimized data layout and larger compression units.

Q67. What happens to dedupe/compression during a host failure?

► **Answer:** During a host failure, vSAN continues operating based on the storage policy. Deduplication and compression remain active on healthy hosts. When the failed host is repaired/replaced, data resyncs and space efficiency is maintained.

Q68. Does enabling dedupe/compression require a rolling reformat?

► **Answer:** Yes, enabling or disabling deduplication and compression in OSA requires an evacuation and reformatting of all disk groups. This is a disruptive process that should be planned with a maintenance window.

Q69. What is vSAN space reporting?

► **Answer:** vSAN reports storage capacity as used capacity, reserved capacity, free capacity, and health reserve. Space efficiency features show actual vs. logical usage. The vSAN Capacity view in vCenter provides detailed breakdown including dedupe/compression savings.

Q70. What is the impact of dedupe/compression on performance?

► **Answer:** Deduplication and compression add CPU overhead but can improve effective I/O throughput by reducing the amount of data written to disk. In most workloads, the performance impact is minimal on modern servers with sufficient CPU resources.

◆ 7. vSAN Operations & Maintenance

15 Questions – Covering key concepts and exam topics

Q71. How do you put a vSAN host into maintenance mode?

► **Answer:** Right-click the host in vCenter > Enter Maintenance Mode. Choose a data evacuation mode: Full Data Migration (evacuates all data), Ensure Accessibility (minimal migration), or No Data Migration. For planned maintenance, Full Data Migration is recommended.

Q72. What are the three maintenance mode options in vSAN?

► **Answer:** 1) Full Data Migration: moves all data off the host, slowest but safest. 2) Ensure Accessibility: only moves data that would become inaccessible, faster. 3) No Data Migration: puts host in maintenance immediately, objects become non-compliant if the host holds critical components.

Q73. How do you add a new host to a vSAN cluster?

► **Answer:** Add the host to the vCenter inventory, configure the vSAN VMkernel adapter with appropriate IP, add storage devices to disk groups, then add the host to the vSAN cluster. vSAN will automatically rebalance data across the new host.

Q74. How do you replace a failed disk in vSAN?

► **Answer:** Mark the failed disk as removed in vCenter, physically replace it, then add the new disk to the disk group in vCenter. vSAN will automatically rebuild data on the new disk from remaining healthy copies.

Q75. What is vSAN rebalancing?

► **Answer:** vSAN rebalancing redistributes storage components across hosts and disk groups to prevent hotspots and ensure even utilization. It is triggered automatically when utilization variance exceeds a threshold or can be initiated manually.

Q76. What is the vSAN proactive rebalance feature?

► **Answer:** Proactive rebalance (vSAN 6.7+) continuously monitors disk utilization and proactively moves components to maintain balance. It runs as a background process with low I/O priority to minimize impact on VM performance.

Q77. How does vSAN handle disk failures?

► **Answer:** When a disk fails, vSAN marks it as absent and begins a 60-minute timer (default). After the timer expires (or if failure is permanent), vSAN initiates component rebuilding using data from surviving copies according to the storage policy.

Q78. What is the repair delay timer in vSAN?

► **Answer:** The repair delay timer (default 60 minutes) is the period vSAN waits before rebuilding components from a failed device. This prevents unnecessary I/O for transient failures (e.g., a host rebooting). The timer can be adjusted in advanced settings.

Q79. How do you remove a host from a vSAN cluster?

► **Answer:** First enter maintenance mode with Full Data Migration to move all data off the host. Verify all objects are compliant. Then remove the host from the vSAN cluster in vCenter and decommission as needed.

Q80. What is vSAN Disk Management?

► **Answer:** vSAN Disk Management in vCenter allows administrators to view disk group status, add/remove disks, mark disks as flash or HDD, and monitor disk health. It provides both GUI and CLI (esxcli vsan storage) interfaces.

Q81. What is the vSAN resync operation?

► **Answer:** Resync rebuilds or re-mirrors vSAN components to restore compliance after a failure, maintenance, or policy change. It uses I/O prioritization to balance resync speed with VM performance. Resync progress is visible in vCenter Monitor view.

Q82. How do you monitor vSAN resync progress?

► **Answer:** Navigate to Cluster > Monitor > vSAN > Resyncing Components in vCenter. This shows all objects being resynced, the bytes remaining, and estimated completion time. CLI: esxcli vsan debug object list.

Q83. What is the vSAN iSCSI target service?

► **Answer:** vSAN iSCSI Target Service allows non-ESXi hosts (Windows, Linux, physical servers) to access vSAN storage as an iSCSI LUN. It is useful for physical workloads that need access to vSAN-backed storage.

Q84. Can you expand a vSAN cluster online?

► **Answer:** Yes, you can add hosts and disk groups to a vSAN cluster while it is online and VMs are running. vSAN will automatically include new capacity and begin rebalancing without requiring downtime.

Q85. What is vSAN Datastore Rescan?

► **Answer:** A datastore rescan updates vCenter's view of the vSAN datastore contents, similar to traditional storage rescans. It ensures vCenter displays the current state of all VMs, VMDKs, and objects stored on the vSAN datastore.

◆ 8. Performance & Monitoring

10 Questions – Covering key concepts and exam topics

Q86. How do you monitor vSAN performance?

► **Answer:** vSAN performance is monitored through vCenter under Cluster/Host > Monitor > vSAN > Performance. It shows IOPS, throughput, latency, and congestion metrics for the cluster, hosts, VMs, and disk groups. Historical data is stored on the vSAN datastore itself.

Q87. What is the vSAN Performance Service?

► **Answer:** The vSAN Performance Service collects and stores performance statistics in a stats database on the vSAN datastore. It must be enabled to view historical performance data. It is enabled per cluster and consumes a small amount of vSAN capacity.

Q88. What metrics does vSAN track?

► **Answer:** vSAN tracks: IOPS (read/write), throughput (MB/s), latency (read/write/recovery), congestion, outstanding I/O, cache hit rate (Hybrid), and resync I/O. Metrics are available at cluster, host, disk group, disk, and VM levels.

Q89. What is vSAN congestion?

► **Answer:** Congestion occurs when the vSAN write buffer (SSD cache in Hybrid, or NVMe write log in ESA) fills faster than data can be destaged. Sustained congestion reduces VM I/O performance. It indicates a need for more cache capacity or higher-tier devices.

Q90. How does vSAN prioritize I/O?

► **Answer:** vSAN uses I/O prioritization to balance VM workload I/O with resync/rebuild I/O. VM I/O gets higher priority by default. The system throttles resync when latency thresholds are exceeded to protect VM performance.

Q91. What is the vSAN cache hit ratio?

► **Answer:** Cache hit ratio measures the percentage of read I/Os served from the SSD cache layer (Hybrid vSAN). A high read cache hit ratio (>90%) indicates effective caching. Low hit ratios suggest the working set is larger than the cache and reads go to HDDs.

Q92. What tools can you use for vSAN performance troubleshooting?

► **Answer:** Tools include: vCenter Performance charts, vSAN Skyline Health, esxtop (for real-time host-level stats), Ruby vSphere Console (RVC), vSAN Observer (legacy web tool), and log analysis via /var/log/vmkernel.log on ESXi hosts.

Q93. What is esxtop and how is it used for vSAN?

► **Answer:** esxtop is a real-time performance monitoring tool run on ESXi hosts via SSH or DCUI. Press 'd' for disk view to see per-device IOPS, latency, and queue depth. Press 'n' for network to verify vSAN VMkernel adapter traffic.

Q94. What is vSAN HCL compliance?

► **Answer:** vSAN requires all hardware (servers, NICs, HBAs, SSDs, HDDs) to be listed on the VMware vSAN HCL (Hardware Compatibility List). Non-HCL hardware is not supported and may cause vSAN health check failures and data integrity issues.

Q95. What is the vSAN Skyline Health?

► **Answer:** vSAN Skyline Health (formerly vSAN Health Check) is an automated health monitoring system in vCenter that checks cluster health across 100+ checks covering hardware, networking, configuration, data, and performance. Results are shown with traffic light severity indicators.

◆ 9. vSAN HCI Mesh & File Services

10 Questions — Covering key concepts and exam topics

Q96. What is vSAN HCI Mesh?

► **Answer:** vSAN HCI Mesh (vSAN 7.0 U1+) disaggregates compute and storage, allowing a compute-only cluster to mount and use storage from a separate vSAN storage cluster over the network. This enables flexible resource scaling.

Q97. What are the network requirements for HCI Mesh?

► **Answer:** HCI Mesh requires 25GbE or higher network connectivity between the compute and storage clusters. The connection uses vSAN over TCP/IP. Low latency is critical; VMware recommends < 2ms RTT between clusters for optimal performance.

Q98. What is vSAN File Services?

► **Answer:** vSAN File Services provides NFS v3/v4.1 shares on top of the vSAN datastore. It allows non-vSphere workloads (Linux, Windows) to access shared file storage. File services are managed through the vSAN File Service agent VMs.

Q99. What is the vSAN File Service agent?

► **Answer:** vSAN File Service agents are small VMs (FSVMs) automatically deployed by vCenter on each host. They expose NFS endpoints and handle file I/O, translating NFS operations to vSAN object store operations. One FSVM is deployed per host by default.

Q100. What protocols does vSAN File Services support?

► **Answer:** vSAN File Services supports NFS v3 and NFS v4.1. NFSv4.1 provides improved security (Kerberos authentication), better performance (session trunking), and enhanced locking semantics compared to NFSv3.

Q101. How is vSAN File Services storage protected?

► **Answer:** Files on vSAN File Services are protected by the vSAN storage policy applied to the file share namespace. Files benefit from vSAN's RAID-1/5/6 protection and are replicated according to the FTT setting, just like regular VM storage.

Q102. What is vSAN Data Persistence Platform?

► **Answer:** vSAN Data Persistence Platform (DPP) is a framework that allows cloud-native stateful applications (like MinIO, Dell EMC PowerScale) to leverage vSAN storage with direct access to vSAN objects via a REST API, bypassing VMDK overhead for maximum performance.

Q103. What is vSAN Max?

► **Answer:** vSAN Max is a disaggregated storage-only offering introduced with vSAN 8 ESA, designed for high-density, petabyte-scale storage clusters. It separates storage from compute for flexibility and supports extremely high capacity NVMe configurations.

Q104. Can a compute cluster mount multiple vSAN storage clusters?

► **Answer:** Yes, with HCI Mesh, a single compute cluster can mount datastores from multiple different vSAN storage clusters simultaneously. This provides flexibility in workload placement and storage tiering across multiple clusters.

Q105. What licensing is required for HCI Mesh?

► **Answer:** HCI Mesh requires vSAN Enterprise or higher license on both the compute cluster hosts and the storage cluster hosts. The vSAN license must include HCI Mesh as a feature. Check the VMware vSAN Feature comparison matrix for current licensing.

◆ 10. vSAN Licensing & Editions

5 Questions — Covering key concepts and exam topics

Q106. What are the vSAN license editions?

► **Answer:** vSAN is available in three main editions: Standard (basic clustering, RAID-5/6, dedup/compression), Advanced (adds Stretched Clustering, iSCSI Target), and Enterprise (adds Data-at-Rest Encryption, HCI Mesh, vSAN Max). vSAN Foundation is a separate entry-level option.

Q107. How is vSAN licensed?

► **Answer:** vSAN is licensed per CPU (processor socket) on ESXi hosts participating in the vSAN cluster. Each physical CPU on each host requires a vSAN license. The license is entered in vCenter's license management.

Q108. What is vSAN Foundation?

► **Answer:** vSAN Foundation (previously vSAN Standard starting vSphere 8) is an entry-level offering bundled with certain vSphere editions. It provides basic vSAN capabilities without advanced features like deduplication, compression, encryption, or stretched clustering.

Q109. Does the vSAN witness host require a license?

► **Answer:** The vSAN witness appliance (VM) does not require a vSAN license. However, the ESXi host running the witness appliance must have a valid vSphere license. The witness is free from a vSAN licensing perspective.

Q110. What is the trial period for vSAN?

► **Answer:** vSAN includes a 60-day evaluation license that provides full Enterprise-level features. After the evaluation period, a valid license must be applied or vSAN will enter a reduced functionality mode (existing VMs remain accessible but new configurations are restricted).

◆ 11. Backup, Recovery & Data Protection

10 Questions — Covering key concepts and exam topics

Q111. How do you back up VMs on vSAN?

► **Answer:** VMs on vSAN can be backed up using any vSphere-compatible backup solution (Veeam, Commvault, Veritas, VMware Live Recovery, etc.) via the VMware vStorage APIs for Data Protection (VADP). vSAN does not provide built-in backup functionality.

Q112. What is VMware Live Recovery?

► **Answer:** VMware Live Recovery (formerly vSphere Replication + Site Recovery Manager) provides VM replication and automated failover/failback between sites. It can replicate VMs from a vSAN cluster to another vSAN cluster or traditional storage.

Q113. Does vSAN have native snapshot support?

► **Answer:** vSAN does not have its own snapshot technology. VM snapshots on vSAN use the native vSphere snapshot mechanism (delta VMDKs). vSAN supports the use of vSphere snapshots but heavy snapshot use can impact performance and should be managed carefully.

Q114. What happens to a VM during a vSAN cluster failure?

► **Answer:** If the vSAN cluster loses quorum (too many components unavailable), the datastore becomes inaccessible and VMs on that datastore will experience I/O errors and potentially crash. VM availability depends on whether the storage policy has sufficient redundancy.

Q115. What is the recovery point objective (RPO) for vSAN stretched clusters?

► **Answer:** vSAN stretched clusters provide zero RPO for site failures because data is synchronously mirrored between sites in real-time. No data is lost when a site fails because the surviving site has up-to-date copies of all VM data.

Q116. Can you replicate from vSAN to non-vSAN storage?

► **Answer:** Yes, vSphere Replication can replicate VMs from a vSAN source to a non-vSAN destination (traditional VMFS or NFS datastore). The destination does not need to be vSAN; it just needs to be a supported vSphere datastore.

Q117. What is vSAN native protection?

► **Answer:** vSAN provides native data protection through its RAID-1/5/6 storage policies. This protects against hardware failures (disk, host failures) but is not a substitute for backup. Native protection does not protect against data corruption, accidental deletion, or ransomware.

Q118. How does vSAN handle VM recovery after a host failure?

► **Answer:** After a host failure, HA (vSphere High Availability) restarts failed VMs on surviving hosts within the cluster. vSAN provides the storage access from remaining healthy hosts. HA restart time depends on HA settings and available cluster resources.

Q119. What is changed block tracking (CBT) in vSAN?

► **Answer:** Changed Block Tracking (CBT) is a vSphere feature (not vSAN-specific) that tracks changed blocks in a VMDK since the last backup. Backup solutions use CBT via VADP to perform incremental backups efficiently, reducing backup windows.

Q120. Can vSAN datastores be used with VMware Cloud Disaster Recovery?

► **Answer:** Yes, VMware Cloud Disaster Recovery (VCDR) can protect VMs stored on vSAN datastores. VCDR provides cloud-based recovery orchestration and allows failover to VMware Cloud on AWS, offering a cost-effective DR solution for vSAN environments.

◆ 12. Troubleshooting & Advanced Topics

15 Questions – Covering key concepts and exam topics

Q121. How do you check vSAN cluster status from CLI?

► **Answer:** Use: 'esxcli vsan cluster get' to view cluster UUID and node membership. 'esxcli vsan storage list' shows disk group configuration. 'esxcli vsan health cluster list' shows health check results. All commands are run from ESXi SSH.

Q122. What is the vSAN Observer?

► **Answer:** vSAN Observer is a web-based tool (accessible via RVC) that provides deep performance and diagnostic data about a vSAN cluster. It collects metrics from all hosts and presents them in a unified dashboard. It has been largely superseded by the built-in vCenter performance charts.

Q123. How do you collect vSAN support logs?

► **Answer:** Use vCenter > Cluster > Actions > Export Support Bundle to collect a vSAN-specific log bundle. Alternatively, generate a VM support bundle from each ESXi host and from vCenter. The bundle includes vSAN logs, health data, and configuration.

Q124. What causes vSAN split-brain?

► **Answer:** Split-brain occurs when a network partition divides the cluster into isolated segments where each segment believes it should own the data. vSAN prevents this through quorum-based component ownership; the majority partition retains access, and minority VMs are paused.

Q125. What is the vSAN UUID?

► **Answer:** The vSAN cluster UUID is a unique identifier for the vSAN cluster, generated when the cluster is first configured. It is embedded in disk group metadata. If the cluster UUID changes (e.g., after misconfiguration), hosts may not recognize existing disk groups.

Q126. How do you recover vSAN data when the cluster is in a critical state?

► **Answer:** Use vSAN Proactive HA to move VMs before failure. If already in a critical state, use 'esxcli vsan debug object list' to see object health. Contact VMware Support for recovery assistance. Never power off all hosts simultaneously—maintain quorum at all times.

Q127. What is the vSAN object repair timer?

► **Answer:** The object repair timer is a configurable delay (default 60 minutes) before vSAN starts rebuilding absent or degraded components. This prevents unnecessary rebuilds for transient failures. Extended absence triggers rebuilding to restore policy compliance.

Q128. What does 'reduced availability' mean in vSAN?

► **Answer:** Reduced availability means a VM's storage objects have fewer redundant copies than the policy requires, but are still accessible. For example, with FTT=1, losing one host puts objects in reduced availability. The VM runs but has no protection against further failures.

Q129. What is stale data in vSAN?

► **Answer:** Stale data occurs when a component's data is outdated compared to the current version (e.g., after a disk or host has been absent). vSAN tracks component versions and rebuilds stale components using fresh data from healthy copies.

Q130. How do you reclaim space in vSAN after deleting VMs?

► **Answer:** vSAN automatically reclaims space when VMs are deleted. For thin-provisioned disks, space is freed immediately. Dead space from snapshots and deleted objects is reclaimed by the CLOM garbage collector. UNMAP/TRIM is supported for SSD capacity reclamation.

Q131. What is vSAN cluster partition?

► **Answer:** A cluster partition occurs when the vSAN cluster is split into two or more isolated groups that cannot communicate. Each partition has some components but not all. vSAN uses the witness to break ties and determine which partition continues operating.

Q132. How does vSAN handle rolling ESXi upgrades?

► **Answer:** vSAN supports rolling upgrades where hosts are patched one at a time while remaining in the cluster. Each host enters maintenance mode, gets upgraded, exits maintenance mode, then the next host is updated. vSAN resyncs data between each host upgrade.

Q133. What is vSAN on-disk format?

► **Answer:** The vSAN on-disk format determines how data is stored on disk groups. Different versions of vSAN introduce new on-disk formats with improvements. Upgrading the on-disk format requires all hosts to be on the new vSAN version and may require disk group reformatting.

Q134. What are vSAN advanced configuration options?

► **Answer:** Advanced configuration options (accessible via vCenter Cluster > Configure > vSAN > Advanced Options) include repair delay timer, large cluster support, software checksum, automatic rebalance threshold, and other performance/behavior tuning parameters.

Q135. What is vSAN Data Evacuation precheck?

► **Answer:** Before entering maintenance mode, vSAN performs a precheck to estimate whether it can fully evacuate data from the host. The precheck considers available space, cluster capacity, and current policy requirements. If insufficient capacity exists, the precheck warns the administrator.

◆ 13. vSAN 8 & ESA Specific Topics

10 Questions — Covering key concepts and exam topics

Q136. What are the key differences between vSAN OSA and ESA?

► **Answer:** vSAN OSA (Original Storage Architecture) uses cache + capacity disk groups with traditional SSD/HDD tiering. vSAN ESA (Express Storage Architecture) uses a single-tier NVMe-only design with a new log-structured storage engine providing better performance, efficiency, and simpler management.

Q137. What hardware is required for vSAN ESA?

► **Answer:** vSAN ESA requires: NVMe storage devices only (no SATA/SAS), NVMe SSDs that are vSAN ESA certified on the HCL, a minimum of 3 NVMe devices per host, ESXi 8.0+, and sufficient memory (VMware recommends 512GB+ RAM for optimal performance).

Q138. Does vSAN ESA support RAID-5/6?

► **Answer:** Yes, vSAN ESA significantly improves RAID-5 and RAID-6 (erasure coding) performance compared to OSA. ESA's optimized write path reduces the write amplification associated with erasure coding, making RAID-5/6 viable for performance-sensitive workloads.

Q139. What compression improvements did ESA introduce?

► **Answer:** vSAN ESA uses a new adaptive compression algorithm that achieves higher compression ratios than OSA's LZ4. ESA applies compression inline during the write path at larger chunk sizes, improving both compression ratio and throughput compared to OSA.

Q140. Can you upgrade from vSAN OSA to ESA?

► **Answer:** Direct in-place upgrade from OSA to ESA is not supported. Migrating to ESA requires a fresh deployment with ESA-certified NVMe hardware, followed by data migration from the OSA cluster. VMware provides guidance for phased migration approaches.

Q141. What is the vSAN ESA snapshot engine?

► **Answer:** vSAN ESA introduces a new native snapshot engine that is more efficient than vSphere snapshots. ESA snapshots use redirect-on-write rather than delta VMDKs, providing better performance and eliminating the performance penalty associated with large snapshot chains.

Q142. What is vSAN Max and how does it differ from standard vSAN?

► **Answer:** vSAN Max is a storage-disaggregated variant of vSAN ESA designed for large-scale storage. It runs as a storage-only cluster (no VMs on storage nodes) and provides NVMe-based storage to compute clusters via HCI Mesh. It supports petabyte-scale deployments.

Q143. What improvements did vSAN 8 U1 bring?

► **Answer:** vSAN 8 U1 introduced improvements including enhanced ESA performance, improved RAID-5/6 efficiency, support for additional NVMe devices, improved vSAN Max capabilities, enhanced Skyline Health checks for ESA, and new PowerCLI cmdlets for ESA management.

Q144. What is the single storage tier in vSAN ESA?

► **Answer:** Unlike OSA which separates cache and capacity tiers, ESA uses a single tier of NVMe storage. All NVMe devices contribute to both performance (write staging) and capacity. The log-structured storage engine manages data layout internally without explicit cache/capacity separation.

Q145. What advanced encryption features does ESA support?

► **Answer:** vSAN ESA supports all existing vSAN encryption features plus additional capabilities. Data-at-Rest Encryption uses AES-256-XTS for improved performance on NVMe. Data-in-Transit Encryption is built-in. ESA also supports credential-free encryption for simplified key management.

◆ 14. PowerCLI & Automation

5 Questions – Covering key concepts and exam topics

Q146. How do you get vSAN cluster information using PowerCLI?

► **Answer:** Connect-VIServer -Server vcenter. Then use: Get-Cluster 'ClusterName' | Get-VsanClusterConfiguration to retrieve vSAN cluster settings. Get-VsanDiskGroup lists all disk groups. Get-VsanStoragePolicy lists all storage policies.

Q147. How do you create a vSAN storage policy with PowerCLI?

► **Answer:** Use New-SpbmStoragePolicy with rule sets. Example: New-SpbmRule -Capability (Get-SpbmCapability 'VSAN.hostFailuresToTolerate') -Value 1 followed by New-SpbmStoragePolicy -Name 'Policy1' -RuleSet (New-SpbmRuleSet -Rule \$rule).

Q148. How do you check vSAN health with PowerCLI?

► **Answer:** Use: Test-VsanClusterHealth -Cluster (Get-Cluster 'ClusterName') to run health checks programmatically. Results include all Skyline Health check statuses. Use -TestId to run specific individual tests.

Q149. How do you configure vSAN with PowerCLI?

► **Answer:** Enable vSAN: Set-VsanClusterConfiguration -Cluster \$cluster -VsanEnabled:\$true. Create disk group: New-VsanDiskGroup -VMHost \$host -SsdCanonicalName 'naa.xxx' -DataDiskCanonicalName 'naa.yyy'. Configure policy: New-SpbmStoragePolicy.

Q150. How do you monitor vSAN performance with PowerCLI?

► **Answer:** Use Get-VsanStat -Cluster \$cluster -Stat 'Performance.DiskGroup.IOPs.Read' to retrieve specific performance metrics. PowerCLI vSAN cmdlets provide access to the same data as the vCenter performance charts but in a scriptable format.

◆ 15. vSAN Certification Exam Specific

15 Questions – Covering key concepts and exam topics

Q151. Which VMware certifications cover vSAN?

► **Answer:** Key certifications include: VCP-DCV (VMware Certified Professional - Data Center Virtualization) which covers vSAN basics, VCAP-DCA (Deploy) and VCAP-DCD (Design) for advanced vSAN knowledge, and VCDX-DCV for expert-level design. There was also a VCP-vSAN certification (now merged).

Q152. What is the minimum FTT and host count for RAID-5?

► **Answer:** RAID-5 (erasure coding) with FTT=1 requires a minimum of 4 hosts. This is because RAID-5 needs $n+1$ data disks where $n=3$, so data is striped across 3 hosts + 1 parity host. This is a common exam question.

Q153. What is the minimum FTT and host count for RAID-6?

► **Answer:** RAID-6 with FTT=2 requires a minimum of 6 hosts. RAID-6 uses $n+2$ parity (two parity drives), requiring at least 4 data segments + 2 parity segments = 6 hosts minimum. This is frequently tested in certification exams.

Q154. What storage policy is used when no policy is explicitly assigned?

► **Answer:** The 'vSAN Default Storage Policy' is applied automatically to VMs deployed on a vSAN datastore without an explicit policy. It uses FTT=1, RAID-1 mirroring, with no striping, no Flash Read Cache Reservation, and thin provisioning.

Q155. How many components does a VMDK with FTT=1 RAID-1 have?

► **Answer:** A VMDK with FTT=1 RAID-1 has 3 components: 2 data components (mirrors on 2 different hosts) and 1 witness component (on a third host). This is the fundamental 3-component layout for basic vSAN mirroring.

Q156. What is the maximum VMDK size on vSAN?

► **Answer:** vSAN supports VMDKs up to 62TB (same as vSphere maximum). Large VMDKs are automatically split into multiple objects of up to 255GB each, each independently protected by the storage policy.

Q157. What vSAN feature requires a KMS server?

► **Answer:** vSAN Data-at-Rest Encryption requires a KMIP 1.1 compatible Key Management Server (KMS). Data-in-Transit Encryption does NOT require a KMS. The KMS stores the Key Encryption Keys (KEKs) used to protect local Data Encryption Keys.

Q158. What does the 'Ensure Accessibility' maintenance mode option do?

► **Answer:** Ensure Accessibility migrates only the minimum number of components necessary to ensure no object becomes completely inaccessible during maintenance. It is faster than Full Data Migration but does not fully protect against reduced redundancy during the maintenance window.

Q159. Which vSAN edition includes encryption?

► **Answer:** vSAN Data-at-Rest Encryption is included in the vSAN Enterprise edition. It is not available in Standard or Advanced editions. Data-in-Transit Encryption is available in all editions from vSAN 6.6+.

Q160. What is the maximum number of disk groups per host in vSAN?

► **Answer:** Each ESXi host in a vSAN cluster can have a maximum of 5 disk groups. Each disk group contains 1 cache device and 1-7 capacity devices. So maximum capacity devices per host = 5 disk groups × 7 capacity devices = 35 capacity devices per host.

Q161. What happens when a vSAN cluster runs out of space?

► **Answer:** When vSAN capacity reaches 100%, new write I/Os will fail with out-of-space errors. VMs may crash or freeze. Reads continue. To resolve: delete unused VMs/snapshots, add hosts/disks, change policies to less redundant options, or enable thin provisioning.

Q162. Does vSAN support thick provisioned virtual disks?

► **Answer:** Yes, vSAN supports thick provisioning through the 'Object Space Reservation' storage policy parameter. Setting OSR to 100% creates a fully thick-provisioned VMDK. However, thin provisioning (OSR=0%) is the default and more efficient use of vSAN capacity.

Q163. What is the vSAN datastore name format?

► **Answer:** The vSAN datastore is automatically named 'vsanDatastore' by default when the cluster is configured. It appears as a single shared datastore accessible to all hosts in the cluster. The name can be renamed through vCenter storage settings.

Q164. How does vSAN handle VM swap files?

► **Answer:** VM swap files (.vswp) on vSAN are assigned a default policy with FTT=1 RAID-1 for protection. However, you can configure a separate swap policy with lower protection (FTT=0) to reduce overhead, accepting the risk that swap space is unprotected.

Q165. What is the purpose of the vSAN health check plugin?

► **Answer:** The vSAN Health Check plugin (integrated into vCenter as Skyline Health) performs automated diagnostics covering hardware compatibility, network configuration, cluster health, data integrity, and performance. It helps proactively identify issues before they cause outages.
